Results And Discussion:-

In our experiment, the real-time detections of EEG signals collected from students during class are analyzed through a system developed in Python to evaluate the cognitive state of such learners. The system utilizes a TGAM module, which is connected through a serial port for recording EEG data, with specific concentration on attention levels and brainwave activities such as alpha, beta, and gamma activities. The dry electrode is placed at the Fp1 site as the active electrode, while the reference electrode will be placed at the A1 site according to the international 10-20 electrode placement system. Data acquisition entails capturing the EEG signals and analysis of the same to arrive at a set of various brain wave bands. The data collected is integrated and updated, and it is displayed using a graphical user interface created using the Tkinter framework. This GUI shows the current attention levels, as well as average attention per day and per month, and uses Matplotlib to plot the various neurophysiological indices. It helps us define the periods spent at rest, working intensely, and engaging in deep info processing by detecting waves of alpha, beta, and gamma bands. Since it incorporates both artificial intelligence and sophisticated algorithms, one can effectively teach based on the student's cognitive level during the learning process; this could prove beneficial to the learning process. This kind of an approach would enable tracking of student's attention and engagement to learning hence enable learning intervention in a personal way.

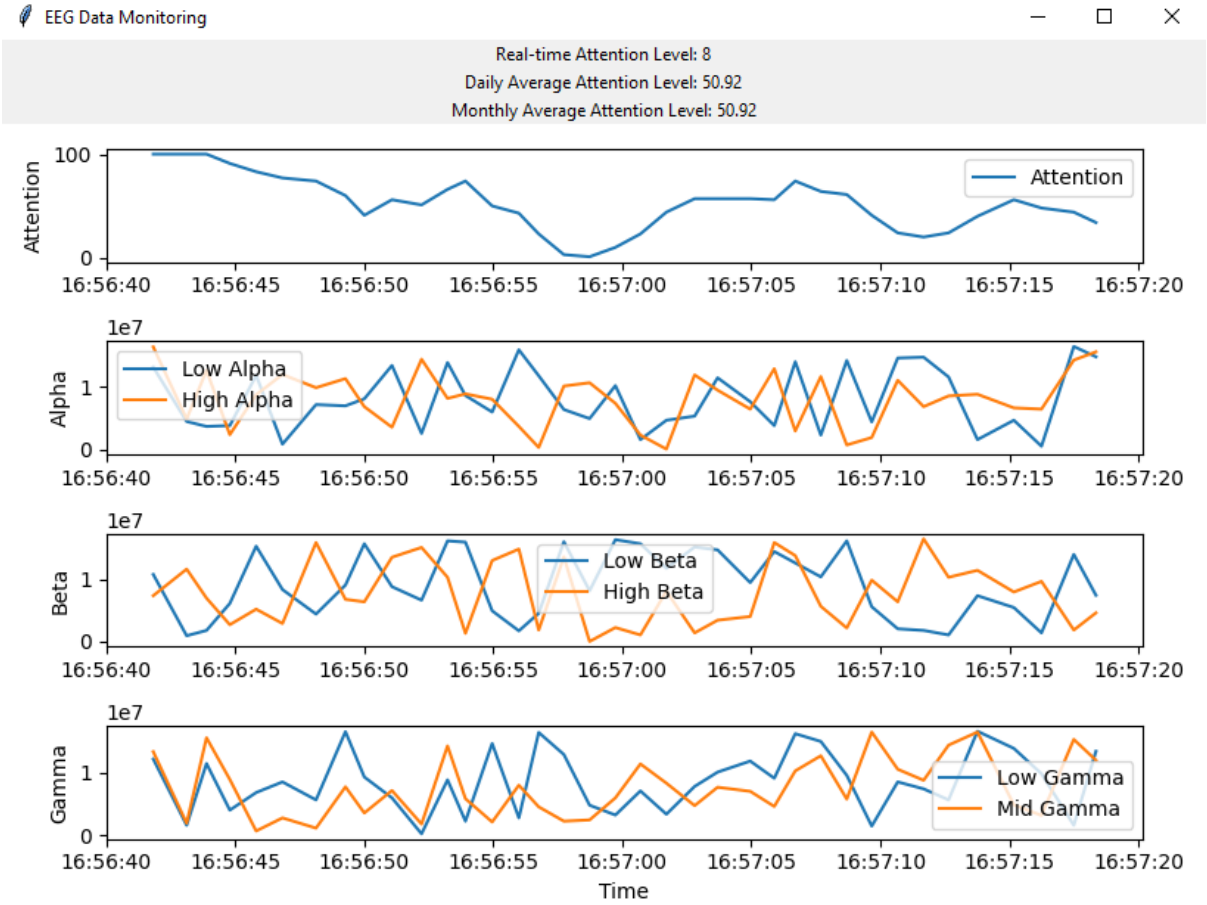


Figure 1: EEG Data Monitoring Visualization

This figure 1 provides a comprehensive overview of EEG data monitoring visualization provides a comprehensive overview of brain activity, segmented into four primary plots: Alertness, illustration, Brain Waves, Alpha Waves, Beta Waves, and Gamma Waves. At the top of the display, the current

attention level is 8 and, average attention level for the day at 50.92 and average attention level for the month 50.92. This stark contrast pencils out the current attention level to be way below the average per-day and average per-month, thus an indicator that the cognitive focus or alertness seems to be low at this moment.

The first graph analyse is about Attention levels with the vertical axis presenting the Attention levels and the horizontal axis indicating the time stamp. The line plot used mimics this trend, but in a downward progression of levels of attention as time elapses. This trend identifies potential time periods where the subject is proven to be unengaged or moderately disconnected, which may be a result of distraction or decreased capability.

The second graph is also on Alpha waves, with Y-AXIS representing the Amplitude and the X-AXIS representing Time-stamps. Two line plots are included: , and for the second one, participants with high levels of alpha waves were selected, individuals with low degree of alpha waves are selected for the second one. Alpha waves are linked to a wakeful state of mind with some amount of idleness, known during periods of meditation or when one is engaged in activities that do not require focusing the brain a lot. The plot shows changes in both low and high alpha waves demonstrating fluctuations in participant's relaxation and extent of preparation for cognitive tasks. High alpha density can be attributed to periods of relaxation in which subjects feel safe and at ease, low alpha density may be interpreted as lowered relaxation, or even elevated levels of arousal.

The third graph shows Beta waves which depicts thinking, concentration and mental activity. Since beta waves are relevant to an awake and active state, the amplitude of beta waves is evident on the Y-axis of the graph whereas the X-axis displays time stamp. The graph is divided in to low beta and high beta waves. While the low level of beta waves, which was previously called Sensorimotor Rhythm (SMR), points to the relaxation maintained with focus, the middle and high levels depict the levels of active learning and wakefulness, correspondingly. The changes in amplitude of beta waves are associated with the changes in the mental load, awareness of surroundings, and attentiveness, whereas the high amplitudes might be related to stress or increased mental activity.

The fourth graph represents Gamma waves that occur during sleep in connection with the brain's maximum productivity and ability to integrate various stimulations. This channel has an attribute of the Y-axis, which refers to the amplitude and the X-axis represents timestamps. This plot distinguishes the low gamma and mid gamma wave discrepancies. Gamma is the highest frequency of brain wave activity; it is involved with high mental activity such as innovative thinking, visual images, and awareness. These fluctuations can be attributed to changes in cognitive workload and the brain's integrative processes and increased gamma waves indicate active thinking and information processing.

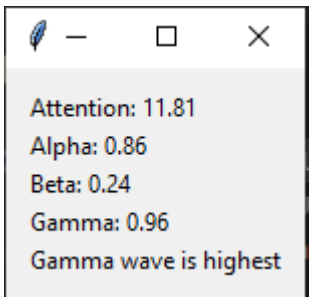
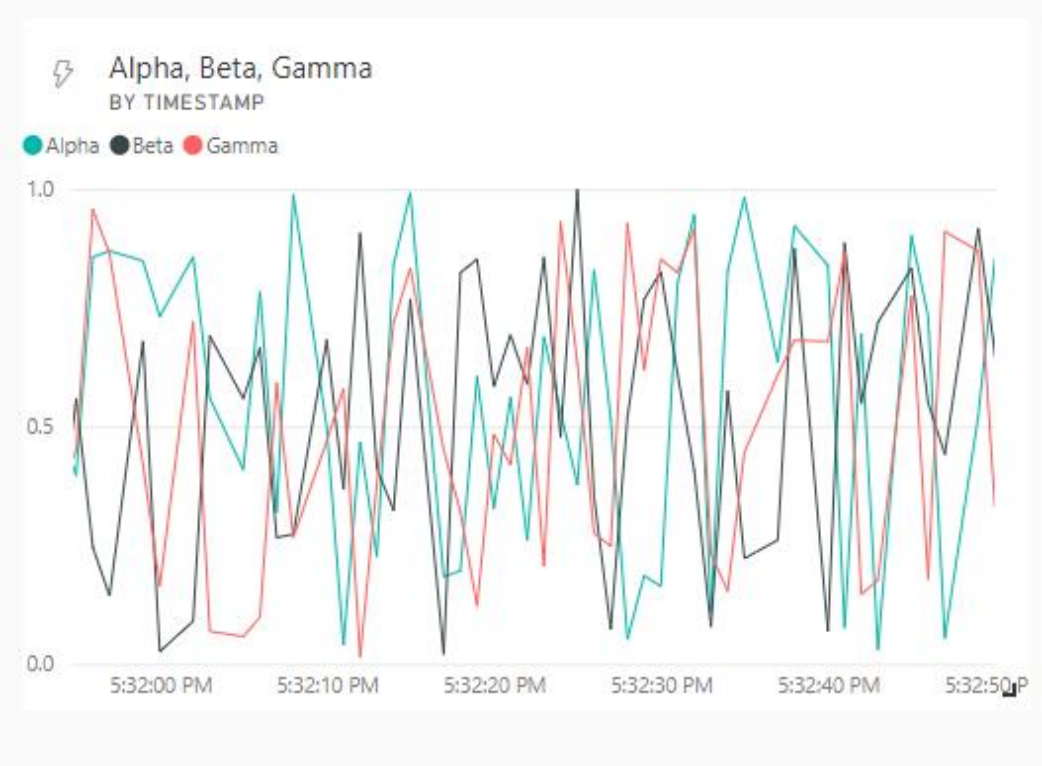


Figure 1.1: Real-time Attention and Brain Waves

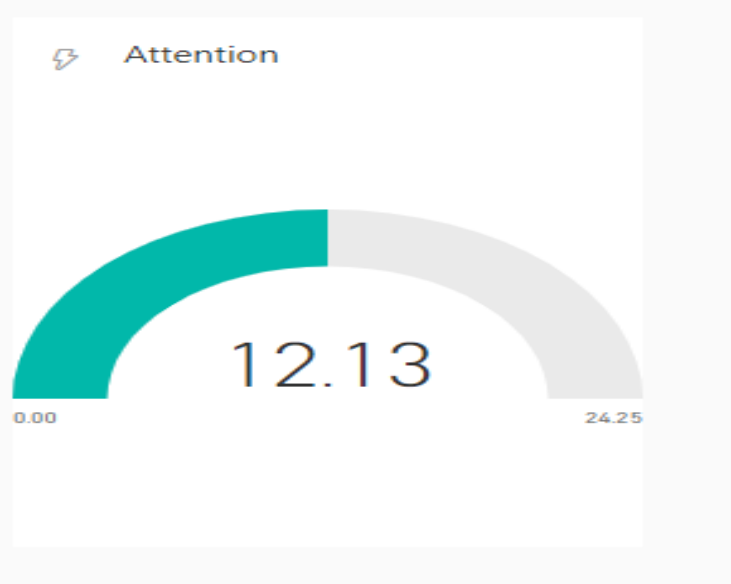
The real-time monitoring label in Figure 1.1 shows the absolute values of attention and brain wave amplitudes at the current time section. This is the case at this snapshot of attention level 11. 81 which is on average, thus inferring that there is moderate focus. From the report, we can determine that the Alpha wave amplitude is 0.86. Beta wave amplitude is 0. 24, and Gamma wave amplitude is 0.96. In particular amongst them, the Gamma wave has the highest amplitude of beating, this indicates a peak in the mental activity and thought process. The assessments done in real-time are vital for the participant as a way of noticing changes in their cognitive condition that may require the need for an intercession to be made instantaneously. For example, in situations that require steady concentration, it is possible to identify slips in one's focus by tracking these values and then readjust to heights of improved performance.



Graph 1: Real-time Alpha, Beta, and Gamma Waves

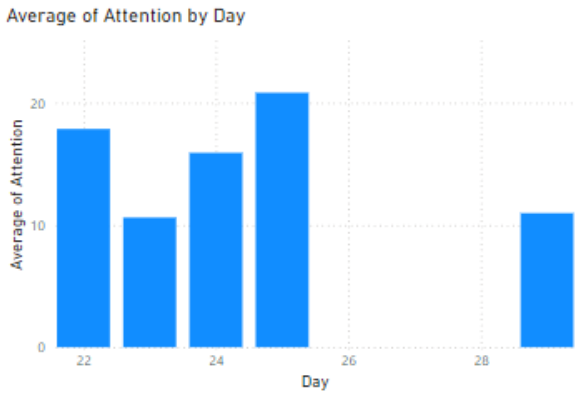
The first graph 1 in the EEG data monitoring visualization of the state is the real-time Alpha, Beta and Gamma oscillations. The Y-axis gives the amplitude of these waves as the X-axis shows the time in seconds. Each wave type is color-coded for clarity: Alpha waves are represented by green, Beta

waves by black while Gamma waves are represented by red. Alpha waves are generally associated with rest and relaxation. Its amplitude fluctuations suggest that they are cycling between wakefulness and sleep-wake cycles. High amplitude in Alpha wave indicates that the subject is either resting or is relaxed while low amplitudes may suggest the subject is changing from rest to arousal. Beta waves which are related to active thinking and attentive focus are also highly variable. Beta waves with high amplitude related with activity and consciousness and low amplitude with the subject being relaxed. High-frequency gamma oscillations, associated with rather complex cognitive processes and increased cortical activity, show significant variability, suggesting periods of increased cognitive and information processing. The tracking of all three wave types at the same time offers a detailed characterization and real-time description of the participant's cognitive and affective state, thus showing how the brain oscillates between different states of relaxation, attention, and load.



Graph 2: Real-time Attention Level

This graph 2 is the gauge chart which is designed to illustrate the real time attention level. The gauge ranges from 0.00 to 24.25, with the current attention level marked at 12.13. This graphical representation can be used for assessing the participant's concentration level at the given moment. An attention level of 12. 13 is not high, which implies that the participant is not highly focused but is not fully distracted either. This real time measure is important for applications that need immediate and constant attention such as during activities that require sustained mental attention. By measuring these attention levels, one is able to notice when focus is low and therefore correct them so that optimum performance is achieved.



Graph 3: Average Attention by Day

Graph 3 is a bar graph that shows the average attention span for the subjects over several days, with the Y-axis representing the average attention and the X-axis showing specific days. From the bar chart we can see changes in the average amount of attention more days showing high average attention like day 26 while other days show low average attention like day 24. Such daily fluctuation shows how attention might be as predictive construct which may be affected by features like sleep, stress, work, and mental health.

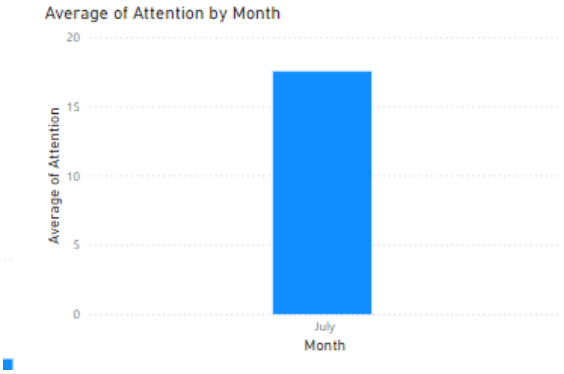


Figure 2: Average Attention by Month

The bar chart in figure 2 above reveals the average attention level obtained over the month of July. The Y-axis shows the average response to attention while the X-axis depicts the month. The value on the y axis represents the average of attention levels combined from all sessions that occurred in July. This gives an average attention score of approximately 15 meaning that the participant was highly attentive the entire month in regard to the material presented in the lessons. This data is beneficial for knowing the trends in performance and understanding when there are spikes or drops in attendance which can be attributed to the increased or declined focus which in turn can be caused by stress levels, workload or other factors.

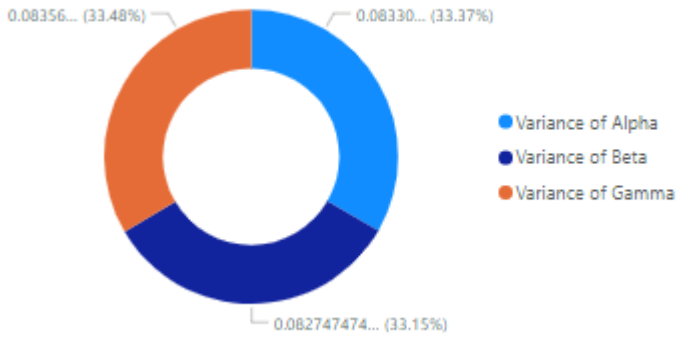


Figure 3: Variance of Alpha, Beta, and Gamma Waves

The donut chart in Figure 3 gives a general idea of how Alpha, Beta, Gamma and other brain waves vary in their oscillations. The chart segments are color-coded: alpha wave is coded blue, beta wave is coded orange, and gamma wave is coded red. The size of each segment equals the variance percentage of the corresponding wave type; with Alpha waves showing a variance of 33.48%, Beta waves at 33.15%, and Gamma waves at 33.37%. This nearly balanced split of variances suggests that the participant undergoes comparable levels of fluctuations in relaxation or Alpha wave activity, focused attention or Beta waves, and high cognitive processing or Gamma waves. This balance has implications of a well functioning and metabolizing brain with no specific dominance by one type of cognition over the others.

